NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Big Data Analysis Metadata 2026/2027

1. Text mining on mental health

mental_health.csv

This dataset contains approximately **43k unique social media text records**. The texts are classified into six emotional or psychological categories: **Positive, Neutral, Anger, Depression, Sadness, and Anxiety**. It is particularly suitable for **text mining, Natural Language Processing (NLP), sentiment/emotion classification, and machine learning**, and could be especially interesting for projects involving **psychology and mental-health-related text analysis**.

Table 1. mental_health.csv data description

Variable name	Description
ID	Unique identifier of the social media text.
Text	Social media text expressing an emotion or psychological state
Emotion	Classification of the text into one of six categories: Positive, Neutral, Anger, Depression, Sadness, or Anxiety

twitter_graph.csv

This dataset contains social network relationship data from Twitter (X), where each row **represents a relationship between a source account and a destination account**. It includes account identifiers, relationship timestamps, account creation dates, follower and following counts, and verification status. The dataset is suitable for data engineering, social network analysis, graph analytics, community detection, etc.

Table 2. twitter_graph.csv data description

Variable name	Description
Source	Identifier of the source Twitter account in the relationship.
Destination	Identifier of the destination Twitter account in the relationship. Together, Source and Destination define a directed edge in the social network.
Conversation ID	Identifier associated with a conversation involving the accounts, where available.
Relationship Date (UTC)	Date and time when the relationship was recorded, expressed in Coordinated Universal Time (UTC).
Source Joined Date	Date when the source account joined Twitter
Source Followers	Number of accounts following the source account.
Source Followed	Number of accounts the source account follows.
Source Verified	Indicates whether the source account is verified (TRUE or FALSE).
Destination Joined Date	Date when the destination account joined Twitter.
Destination Followers	Number of accounts following the destination account.
Destination Followed	Number of accounts the destination account follows.

Destination Verified	Indicates whether the destination account is verified (TRUE or FALSE).

2. Asian restaurant sales

asian_restaurant.csv

This dataset contains transaction-level data from an Asian restaurant chain, where each row represents an **invoice line**. It combines information about products, sales, customers, employees, and orders, making it suitable for **data engineering, exploratory analysis, customer analytics, sales forecasting, and machine learning**. Since each invoice can contain multiple product lines, the dataset also provides opportunities to analyse **purchasing patterns and customer behaviour**.

Table 3. asian_restaurant.csv data description

Variable name	Description
DocNumber	Invoice/document number. The same number can appear across multiple rows belonging to the same invoice.
ProductDesignation	Name or designation of the product sold.
ProductFamily	Product family to which the product belongs. Each product belongs to one family.
Qty	Quantity of the product sold.
TotalAmount	Total sales amount for the quantity purchased.
InvoiceDateHour	Date and time when the invoice was issued.
EmployeeID	Identifier of the employee who issued the invoice.
IsDelivery	Indicates whether the order was a delivery (1) or dine-in (0).
Pax	Number of people at the table.
CustomerID	Identifier of the customer, when a customer record is associated with the sale.
CustomerCity	City of the customer, mainly relevant for delivery orders.
CustumerSince	Date when the customer record was created.

europe_air_routes.csv

This dataset contains flight route information between major airports in Europe, where each **row represents a route between a departure airport and an arrival airport**. It combines airport information, geographical coordinates, flight schedules, estimated prices, flight durations, and airline operator details. The dataset is suitable for flight network analysis, and machine learning.

Table 4. europe_air_routes.csv data description

Variable name	Description
id	Unique identifier of the flight route.
iata_from	IATA code of the departure airport.
iata_to	IATA code of the arrival airport.
price	Listed or estimated price associated with the route. The currency should be confirmed from the original data source.
day1	Indicates whether flights are scheduled on Monday.
day2	Indicates whether flights are scheduled on Tuesday.
day3	Indicates whether flights are scheduled on Wednesday.

day4	Indicates whether flights are scheduled on Thursday.
day5	Indicates whether flights are scheduled on Friday.
day6	Indicates whether flights are scheduled on Saturday.
day7	Indicates whether flights are scheduled on Sunday.
flights_per_day	Approximate frequency of flights per day on the route.
common_duration	Typical flight duration for the route, expressed in minutes.
flights_per_week	Number of flights scheduled per week on the route.
first_flight	Date of the first planned flight for the route.
last_flight	Date of the last planned flight for the route.
airlineroutes	List of airline and route records, potentially including airline names, carrier codes, aircraft codes, and other operator information. Stored as a nested structure.
arrival_airport_country_code	Country code associated with the arrival airport.
arrival_airport_name	Name of the arrival airport.
arrival_airport_city_name	City associated with the arrival airport.
arrival_airport_city_name_en	English-language name of the arrival airport's city.
arrival_airport_country	Country in which the arrival airport is located.
arrival_airport_no_routes	Number of routes associated with the arrival airport, according to the dataset.
departure_city	City associated with the departure airport.
departure_country	Country in which the departure airport is located.
departure_ICAO	ICAO code identifying the departure airport.
departure_latitude	Latitude of the departure airport.
departure_longitude	Longitude of the departure airport.
departure_altitude	Altitude of the departure airport, typically expressed in feet.
arrival_ICAO	ICAO code identifying the arrival airport.
arrival_latitude	Latitude of the arrival airport.
arrival_longitude	Longitude of the arrival airport.
arrival_altitude	Altitude of the arrival airport, typically expressed in feet.

3. Churn prediction

churn.csv

This dataset contains customer-level information from a bank, including demographic characteristics, financial information, account activity, and the products and services used by each customer. The main objective is to understand and predict **customer churn**, represented by the *Exited* variable, which indicates whether a customer left the bank. This makes the dataset particularly suitable for **classification, customer segmentation, churn prediction, and customer behaviour analysis**.

Table 5. churn.csv data description

Variable name	Description
RowNumber	Row number in the dataset.
CustomerId	Internal identifier of the customer.
Surname	Customer's surname.
CreditScore	Customer's credit score; higher values indicate a better credit rating.
Geography	Customer's location/country.
Gender	Customer's gender.
Age	Customer's age.

Tenure	Number of years the customer has been with the bank.
Balance	Total balance held by the customer across their accounts.
NumOfProducts	Number of banking products held by the customer.
HasCrCard	Indicates whether the customer has a credit card (1: Yes, 0: No).
IsActiveMember	Indicates whether the customer is an active bank member (1: Yes, 0: No).
EstimatedSalary	Estimated annual salary of the customer.
Exited	Indicates whether the customer left the bank (1: Yes, 0: No).

bank_transactions.csv

This dataset contains transaction-level financial data, where **each row represents a financial transaction between a sender account and a receiver account**. It combines transaction details, account identifiers, timestamps, fraud indicators, geographical information, device usage, and network performance metrics. The dataset is suitable for financial network analysis by analysing transaction networks, identify suspicious account relationships, detect potential fraud patterns, and investigate how network performance relates to transaction outcomes.

Table 6. bank_transactions.csv data description

Variable name	Description
Transaction ID	Unique identifier for each transaction, ensuring traceability.
Sender Account ID	Identifier of the account initiating the transaction.
Receiver Account ID	Identifier of the account receiving the transaction.
Transaction Amount	Monetary value transferred or involved in the transaction.
Transaction Type	Type of transaction, such as Transfer, Withdrawal, or Deposit.
Timestamp	Date and time when the transaction occurred.
Transaction Status	Indicates whether the transaction was successful or failed.
Fraud Flag	Binary indicator identifying whether the transaction was flagged as potentially fraudulent.
Geolocation (Latitude/Longitude)	Geographic coordinates associated with the transaction, enabling spatial analysis.
Device Used	Type of device used to conduct the transaction, such as Mobile or Desktop.
Network Slice ID	Identifier of the network slice used to process the transaction.
Latency (ms)	Transaction-related network delay, measured in milliseconds.
Slice Bandwidth (Mbps)	Available bandwidth of the network slice, measured in megabits per second.
PIN Code	Four-digit security code associated with the transaction, masked for privacy.